



SM Shaiful Islam Tutul

EEE Researcher & Embedded Systems Engineer

Profile

Dedicated 2nd-year Electrical and Electronic Engineering student at Jamalpur Science And Technology University (JSTU) specializing in Hardware Spiking Neural Networks (SNN), Neuromorphic Computing, Embedded Systems, and Biomedical Telemetry. Passionate about brain-inspired silicon, analog sensor loops, and building autonomous hardware prototypes.

Work Experience

2024 — Present

Department of EEE, JSTU | Jamalpur, Bangladesh

Hardware SNN & Embedded Systems Researcher

- Leading architectural research on hardware Spiking Neural Networks (SNN) and event-driven bio-inspired computing paradigms.
- Designing and prototyping a hardware SNN-based RC Vehicle Controller for reflexive sub-5ms low-power mobile robotics.
- Architected ESP32-based Smart E-Voting System with cryptographic audit logs and biometric security.

2024 — 2025

ROBOFUSION 1.0 — National Robotics Festival

Lead Embedded Systems Engineer — Project Clever Sapiens

- Engineered real-time ECG monitoring station integrating AD8232 sensor and Arduino Nano with serial synchronization.
- Built 60fps multithreaded Python GUI with R-wave BPM detection and automated clinical-standard WhatsApp PDF report delivery.
- Earned official Certificate of Participation for demonstration excellence at the national robotics festival.

2023 — 2024

Academic & Hardware Robotics Labs | JSTU

Autonomous Robotics & IoT Developer

- Developed multi-sensor autonomous fire-fighting robot with flame sensor array and servo pump targeting.
- Conducted spectral analysis, bio-signal filtration, and scientific simulations using Python (SciPy/NumPy) and Brian2.

Contact

smshaifulislam46@gmail.com

+880 1798-955785

JSTU, Jamalpur, Bangladesh

github.com/smsitutul

linkedin.com/in/smsitutul

Education

B.Sc. in Electrical & Electronic Eng.

Jamalpur Sci. & Tech. Univ. (JSTU)

2nd Year (2023 – 2027)

Higher Secondary Certificate (HSC)

Guthia Ideal College

Completed | GPA 4.50 / 5.00

Secondary School Certificate (SSC)

Narayanpur Palli Union Inst.

Completed | GPA 5.00 / 5.00

Skill

- Hardware SNN & Brain2
- Neuromorphic Architectures
- Embedded C/C++ & Python
- Biomedical DSP (AD8232)
- Arduino, ESP32 & STM32
- Circuit Design & Simulation
- Real-Time Telemetry & GUI
- Scientific Stack (NumPy/SciPy)

Awards

ROBOFUSION 1.0

National Robotics Fest (Participation)

Grameenphone Academy

Personal Branding in AI Era (2026)

Google Cloud & Simplilearn

Image Captioning & GenAI Studio (2026)

Reference

Department of EEE

Jamalpur Sci. & Tech. Univ. (JSTU)

Phone: +880 1798-955785

Email: info@jstu.edu.bd

Academic References

Available Upon Request

Email: smshaifulislam46@gmail.com

GitHub: github.com/smsitutul